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Course: 21CSC202J - Operating Systems

Unit – 2 : Session – 13 : SLO – 1 | Bounded Buffer (Producer-Consumer) — Missing Pseudocode

Answers

TODO 1: wait(empty)

TODO 2: wait(mutex)

TODO 3: signal(mutex)

TODO 4: signal(full)

TODO 5: wait(full)

TODO 6: wait(mutex)

TODO 7: signal(mutex)

TODO 8: signal(empty)

Completed pseudocode

```
semaphore mutex = 1   // mutual exclusion
semaphore empty = N   // empty slots
semaphore full  = 0   // full slots
buffer[N]
```

```
procedure PRODUCER():
    while true:
        item = produce_item()
        wait(empty)
        wait(mutex)
        add_to_buffer(item)
        signal(mutex)
        signal(full)

procedure CONSUMER():
    while true:
        wait(full)
        wait(mutex)
        item = remove_from_buffer()
        signal(mutex)
        signal(empty)
        consume_item(item)
```